

Jyot Antani, PhD

jyot.antani@yale.edu • Twitter: [@AntaniJo](https://twitter.com/AntaniJo)

Experience Highlights: Biophysics, Microbiology, Virology, Microscopy, Image Analysis

Experience & Education

Jun 2021- Now	Postdoctoral Research Associate, Yale University (USA) Paul Turner Lab , Department of Ecology and Evolutionary Biology Thierry Emonet Lab , Molecular, Cellular, and Developmental Biology Research: Viral target search on host surfaces - Dynamics, Evolution, and Therapeutic Application See proposal for details.
Aug 2016- May 2021	PhD, Pushkar Lele Lab , Department of Chemical Engineering, Texas A&M University (USA) Dissertation title: Sensory Functions of the Bacterial Flagellar Motor
Jul 2012- May 2016	Bachelor of Technology with Honours in Chemical Engineering Indian Institute of Technology Bombay (IIT Bombay), Mumbai (India)

Publications

- [Antani JD](#), Gupta R, Lee AH, Rhee KY, Manson MD, Lele PP (2021) Mechanosensitive recruitment of stator units promotes binding of the response regulator CheY-P to the flagellar motor. *Nature Communications*. doi: [10.1038/s41467-021-25774-2](https://doi.org/10.1038/s41467-021-25774-2)
- [Antani JD](#), Sumali AX, Lele TP, Lele PP (2021) Asymmetric random walks reveal that the chemotaxis network modulates flagellar rotational bias in *Helicobacter pylori*. *eLife*. doi:[10.7554/eLife.63936](https://doi.org/10.7554/eLife.63936)
- Katiyar A*, [Antani JD](#)*, McKee BP, Gupta R, Lele PP, Lele TP (2021) A method for direct imaging of x-z cross-sections of fluorescent samples. *Journal of Microscopy*. doi:[10.1111/jmi.12965](https://doi.org/10.1111/jmi.12965) (*equal contribution)
- Wong GCL, [Antani JD](#), et al. (2021) Roadmap on emerging concepts in the physical biology of bacterial biofilms: from surface sensing to community formation. *Physical Biology*. doi:[10.1088/1478-3975/abdc0e](https://doi.org/10.1088/1478-3975/abdc0e)
- Ford KM, [Antani JD](#), Nagarajan A, Johnson MM, Lele PP (2018) Switching and torque generation in swarming cells of *E. coli*. *Frontiers in Microbiology*, 9, 2197. doi:[10.3389/fmicb.2018.02197](https://doi.org/10.3389/fmicb.2018.02197)
- [Antani JD](#), Gupta R, Lele PP. The chemotaxis conundrum for a polar-flagellated bacterium. *In preparation*.
- Katiyar A, Zhang J, [Antani JD](#), Yu Y, Lele PP, Reinhart-King CA, Sniadecki NJ, Roux KJ, Dickinson RB, Lele TP. Drop-like nuclear deformation facilitated by lamin A/C allows cells to move through obstacle. *In revision at Journal of Cell Biology*.

Conference Presentations

- [Antani JD](#), Lele PP. Modulation of chemotactic signaling in bacteria by mechanical forces.
Mar 2019 | ChEGSA Annual Symposium, Texas A&M University, College Station (USA).
Oct 2018 | American Institute of Chemical Engineers (AIChE) Annual Meeting, Pittsburgh (USA).
Dec 2017 | Texas Branch of American Society for Microbiology, 2017 Fall Meeting, College Station (USA).
- [Antani JD](#), Lele PP. Asymmetric Swimming and Chemotaxis in *Helicobacter pylori*.
Jan 2021 | Bacterial Locomotion And Signal Transduction (BLAST) XVI Meeting, Zoom (online).
Feb 2020 | Biophysical Society (BPS) Annual Meeting, San Diego (USA).
Jan 2019 | Bacterial Locomotion And Signal Transduction (BLAST) XV Meeting, New Orleans (USA).
- [Antani JD](#), Venkatesh KV. Analysis of Glucose Homeostasis in Humans.
Nov 2015 | Futuristic Approach for Alternatives to Animal Testing [CEFIPRA](#) seminar, IIT Bombay, India.

Honors & Awards

- 2020 CIRT Associate Certificate, Center for the Integration of Research, Teaching, and Learning
2015 Best Poster Award, *AMAT* seminar, IIT Bombay
2015 Undergraduate Research Award URA01, IIT Bombay
2012 All India Rank 1427 in Joint Entrance Examination for engineering, among 479,651 participants
2008 National Talent Search Scholarship, NCERT, India

Technical Skills

Perfusion & Thermal Control Experiments Prototyping & 3D-printing Microscopy: Dark-field, TIRF, FRET,
Bacteria & Phage Culturing Molecular Biology Optical Tweezers, Image Analysis

Computer Proficiency / Computational Skills

Image Processing & Analysis: MATLAB, Python, ImageJ – {Highlight: particle-tracking algorithms}
Design & Development: SolidWorks, AutoCAD, LabVIEW, Molecular Biology {SnapGene, Benchling}
Simulation Packages: ANSYS Fluent, Aspen Plus, OpenFOAM, COMSOL, CHARMM, VMD
Programming: MATLAB, Python, C, C++, Fortran, HTML

Doctoral Research

Molecular mechanisms of host substrate sensing by bacterial flagellar motors

Determined the effect of mechanical forces on chemotactic output of the bacterial flagellar motor, with phase microscopy, optical tweezers, and TIRF imaging. Developed a biophysical model for the regulation of allosteric interactions by mechanical force.

Motility, chemotaxis, and random spread of *Helicobacter pylori*, the carcinogenic pathogen

Devised a technique to experimentally determine the chemotactic activity of *H. pylori*. Studied the biophysical principles governing chemotactic migration by this bacterial species which employs run-reversal motility – different from the traditionally studied *E. coli*.

Design and development of experimental and analytical framework

Designing 3D-printed imaging chambers for specific requirements in microscopy experiments, e.g., perfusion, imaging side-views.
Writing MATLAB-based image analysis codes for tracking motile bacteria, mammalian cell-nuclei, and fluorophores.

Additional Research Experience

Jul 2015- Analysis of Glucose Homeostasis in Humans

Feb 2016 *Guide: Prof. K. V. Venkatesh, IIT Bombay*

Formulated a Hill function-based empirical biochemical model of body glucose concentration following a fixed food-intake, obtained least-squares fits to data acquired from 150+ individuals. Constructed a classification tree based on fit-parameters using machine learning in MATLAB; established a statistical grouping among Diabetic and Healthy individuals.

May-Jul A Novel Protocol for Coating Lipid Bilayers onto Silver Nanocubes

2015 *Guide: Dr. Hung-Jen Wu, Texas A&M University*

Established a protocol to use a tip-sonicator for making small unilamellar vesicles of lipids, to coat lipid bilayers onto Silver nanocubes. Employed a Localized Surface Plasmon Resonance-based nanocube sensor to monitor protein binding on supported lipid bilayers. Analyzed binding kinetics of GM1 ganglioside with Cholera Toxin Subunit B.

May 2014- Design of a Chemical Vapor Deposition Reactor

Apr 2015 *Guide: Prof. Mahesh S. Tirumkudulu, IIT Bombay*

Simulated fluid flow around a disk, successfully obtained the pressure coefficients observed experimentally in literature. Proposed an unconventional design for uniform coating of parylene polymer onto silicon micro-wafers in a micro-scale Chemical Vapor Deposition reactor. Modeled Parylene flow using ANSYS Fluent, enhanced the proposed design.

Jun 2013 Process Improvement and Technology in Jaggery-Making

Guide: Prof. Sanjay M. Mahajani, IIT Bombay

Examined units under development for continuous production as well as recuperation in manufacturing Jaggery, the traditional non-centrifugal cane sugar. Analyzed Jaggery samples for acidity, moisture content, water activity, density, sugar content etc; observed the XRD and ESEM techniques.

Key Course Projects

Spring 2018	Determining Diffusion Coefficient of Microparticles using Particle-Tracking Algorithms <i>Course: Colloidal Applications and Biomolecular Interactions in Life Science</i> Employed particle-tracking algorithms in MATLAB to track the Brownian diffusion of submicron-sized particles recorded using phase-contrast microscopy. Determined the diffusion coefficient of the particles by calculating the average mean-squared displacement of an ensemble.
Spring 2017	Project Proposal: Evolution of Bacterial Flagellum <i>Course: Evolutionary Bioinformatics</i> Proposed a study involving the comparison of flagellar genome across multiple bacterial species which has potential to aid in obstruction of pathogenic motility. Carried out BLAST alignment analysis for nucleotide and protein sequences of flagellin (flagellar filament) gene as preliminary results.
Spring 2016	Review: How Deep a Cell Can Sense <i>Course: Cell Mechanics</i> Carried out a literature survey and reported theoretical, in-vitro as well as in-silico studies on quantified investigations about the range of mechanical signaling.
Fall 2015	On-line Estimation of Concentrations in a Bio-Reactor <i>Course: State Estimation</i> Generated true states from a model based on substrate uptake, phosphorus repression and effect of seed culture on glyco-peptide antibiotic production. Implemented the Simple, Extended and Unscented versions of the Kalman Filter to estimate the concentrations.

Teaching & Mentoring Experience

Fall 2019	A hands-on workshop for image processing and single particle tracking in MATLAB <i>Pushkar Lele Lab, Texas A&M University</i>
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Undergraduate students mentored

Oct 2021- Now	Austin Theroux, Ecology & Evolutionary Biology, Yale University Studying the evolution of motile bacteria against a bacteriophage that attaches through the flagellum
Dec 2019- Jan 2021	Anita Sumali, Biomedical Engineering, Texas A&M University MATLAB-based quantitative analysis of the run-reversal motility of <i>H. pylori</i> ; co-author on Antani et al., eLife, 2021
Apr 2017- Apr 2018	AggiE_Challenge – Undergraduate Mentoring, Texas A&M University Graduate-mentor to a multidisciplinary team of eight undergraduate students employing 3D-printing technology to design scaled-up bacterial models, in order to study the hydrodynamic effects of cell morphology on motility

Teaching Assistant at Texas A&M Chemical Engineering

Spring 2019	Heat Transfer Operations
Spring 2017	Bioprocess Engineering
Fall 2016	Bioprocess Engineering

Teaching, Mentoring, & Management Training

Sep 2021- Dec 2021	Auburn University's Preparing Future Faculty Workshop Interactive workshops consisting panel discussions from faculty and university administrators
Sep 2021	Fundamentals of Inclusive Teaching <i>Poorvu Center: Graduate and Postdoctoral Teaching Development, Yale University</i> A 4-part workshop on creating a respectful, supportive, equitable, and transparent classroom environment
Aug 2018- Dec 2019	Academy for Future Faculty (AFF) , Center for Teaching Excellence, Texas A&M University <i>Faculty Mentor: Dr. Micah Green, Texas A&M University</i> A program designed for training graduate students as future faculty members, featuring weekly seminars and workshops, and anchored by faculty mentorship

Jul- Aug 2019	An Introduction to Evidence-Based Undergraduate STEM Teaching <i>Muti-university Massive Open Online Course from the NSF-founded CIRTl network</i> Eight-week online course on edX.org catering introduction to effective teaching strategies and the research that supports them. Attended local learning community meetings for the course at Texas A&M University.
Oct 2019	Improving Research Mentoring Relationships – Introduction to Effective Mentorship <i>Center for the Improvement of Mentored Experiences in Research (CIMER)</i> 1-day workshop at Texas A&M University, conducted by CIMER from University of Wisconsin-Madison

Leadership & Management Experience

Now	Communications Manager, Yale Postdoctoral Association
Jun 2021	People Management Skills – Course, Chartered Institute of Personnel and Development, UK
Oct 2018- Aug 2020	Secretary (2018-19) & Chair (2019-20), Chemical Engineering Graduate Student Advisory Council <i>Artie McFerrin Department of Chemical Engineering, Texas A&M University</i> Designed the council by-laws as a founding member. Set up channels of communication through surveys and anonymous feedback mechanisms, between grad students and administration in the department. Addressed grad student welfare issues including salary, benefits, mental well-being, work-life balance, student-advisor relationships, facilities, career events, and social events.
Apr 2014- Mar 2015	Competitions Manager, AZeotropy 2015 <i>IIT Bombay's Intercollegiate Chemical Engineering Symposium</i> Pioneered a new model-making competition <i>ChemETimer</i> based on time-precision. Designed a presentation-based competition <i>Swaasthya</i> on engineering-applications to healthcare. Designed <i>Cipher</i>, the online cryptic hunt which received a participation of 400+. Actualized unconventional chemical engineering quizzing for <i>Chem-O-Philia</i>, national level quiz.

COVID-19 Volunteering

2020	Coronasurveys: Monitoring the Incidence of COVID-19 via Open Surveys Volunteered services as the primary translator of the survey to the Gujarati language.
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